Islem Fatma Gamha

AI & Data Engineering MSc Student | End-of-Studies Internship (March 2026)

islem.gamha@dauphine.eu /in/fatma-gamha/ ifatmagamha https://ifg-myportfolio.vercel.app/

Profile

Paris Dauphine–PSL MSc student focused on building data-driven solutions to real-world problems. I have worked on automated data pipelines, NLP-based systems, and ML models from experimentation to deployment. Currently seeking an end-of-studies internship (starting March 2026) in an impact-driven environment.

Education

University of Paris Dauphine - PSL, 09/2024 – Present
Master's in Artificial Intelligence and Big Data

Faculty of Sciences of Tunis, 09/2021 – 06/2024
Bachelor's in Software Engineering and Information Systems

Professional Experience

AI / Data engineering intern, *Carbon Jar* 07/2025 – 09/2025

- Designed and deployed automated data pipelines, integrating multi-source emission factor datasets into a unified database.
- Consolidated emission factors across 40+ regions and 190+ activity types.
- Improved data ingestion efficiency by 30% through modular Airflow DAG design and workflow parallelization.
- Built validation and anomaly detection routines, cutting data quality incidents by 25%.

Skills: Apache Airflow 3.0, ETL, PostgreSQL, Snowflake, Data quality, Workflow optimization, Python, Environmental data, Agentic LCA

Junior Data Analyst, *Insight Research* 11/2024 – 08/2025

- Performed statistical analyses on user behavior and choices datasets to identify key usage patterns.
- Delivered insights that guided feature prioritization and data-driven roadmap decisions.

Skills: R, Statistical tests, Data analysis, Data visualization

Data Scientist / Developer, *Manitoo* 02/2024 – 06/2024

- Designed an automated CV parsing pipeline combining OCR, CNN-based NER and LLMs for semantic normalization and data validation, achieving an F1-score of 86%.
- Deployed the module as a Flask API integrated into a cross-platform HR application, enforcing schema consistency and reducing manual data entry by 40%.

Skills: OCR, ML, CNN (ResNet50), PyTorch, NLP, LLMs, ORM, SQL, GitLab, Flask

Projects

Moodsync, *Hackathon - Hack for Good (Expensya & Université Dauphine) "Coup de Cœur du Jury" Award*
AI-powered therapy assistant designed to enhance patient engagement and optimize therapeutic processes.
Technologies: Python, Streamlit, STT, NLP, Sentiment Analysis, Agents orchestration, LangChain, LangGraph

Predicting profitable startups
Ranking startups based on their profitability potential and predicting their performance to guide investment decisions.

Technologies: ML regression models, Data Analysis, Python, Flask, Fuzzy decision

Alzheimer's Disease Detection
Built a handwriting-based Alzheimer's classifier ranked Top-5 on Kaggle, improving generalization by managing the bias–variance tradeoff.

Tools: EDA, PCA, Feature Engineering, Deep Learning, Ensemble Methods, XGBoost, MLP, Python, Scikit-learn, Pandas, Pytorch

Personalized Recommendation System
Cloud-native movie recommender on GCP using BigQuery and Cloud Run for scalable inference of hybrid CF/CBF recommendation models.
Technologies: GCP (BigQuery, Cloud Run, Cloud Logging), Docker, Python, Recommendation Systems, Hybrid Models

Skills

Programming & Software Engineering
Python (OOP), Java, C, SQL, Git, API design

Machine Learning & Generative AI
NLP, LLMs, Transformers, Embeddings, PyTorch, RAG systems, AI Agents, Reinforcement Learning

Theoretical Foundations
Statistical Learning and optimization, Knowledge Graphs, Decision Theory under uncertainty, Transformers and attention mechanisms

Data Engineering and Big Data
Workflow automation, Data acquisition, Data preprocessing, Distributed processing (Spark, Hadoop), Kafka, Data warehousing

Cloud Tools
Docker, GCP, Git/GitHub, CI/CD, Model monitoring

Languages
French (Fluent), English (Fluent)

Community Involvement

Optima Junior Entreprise, *Project Coordinator & Product Owner* 08/2023 – 08/2024
Managed **end-to-end client projects** (prospecting to delivery), ensuring alignment between requirements and outputs, which enhanced client satisfaction and delivery consistency.

Enactus, *Project & Research Division* 10/2021 – 10/2023
Designed and researched **ecological and social innovation initiatives**, producing proposals that advanced sustainability goals and fostered measurable community impact.